Improving the Biological Validity of Sensory Coding Models through a Neural Spiking Network

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***Abstract*—This work builds upon the locally competitive algorithm (LCA) for sparse neural coding and modeling inhibitory interneuron dynamics. LCA expresses the network dynamics as nonlinear ordinary differential equation and solves for the sparse coefficients for a library of receptive fields. The LCA dynamics in previous work were inspired by properties of neural systems such as the charging of the membrane, but does not incorporate full spiking dynamics. The nodes of the prior LCA models communicate through rate coding but in our model communication between nodes is directly spike based and the asymptotic spike rates of each node are used as the coefficients.**

***Key words*—spike, visual, coding, sparse**

I. Introduction

The inhibitory interneuron population in the cortex is a critical component in cortex activity [1]. However, most theoretical frameworks do not include models for inhibitory populations that are complex enough to approach those found in nature. Such models are unable to faithfully elucidate the role of inhibition in neural coding. Recognizing the need for more robust inhibitory models, Zhu et. al. have developed a framework that incorporates three major properties of inhibitory interneuron populations: (1) respecting Dale's law, (2) maintaining consistent E:I ratios, and (3) allowing for diverse tuning characteristics [2]. Although their work represents a major step in the direction of illuminating the computational function of inhibitory interneurons, they utilize a dynamical system-based model for neurons that directly outputs a firing rate. As an extension of their work and to more closely align with neuronal biophysics, we propose the use of a spiking model for the neurons within their framework. We expect that the behavior of such a model should closely align with previous results using a dynamical system-based framework.

II. Methods

A. Baseline model

A spiking neural network model of the following form was implemented [3]:

$$\tau_m \frac{\mathrm{d}v_i}{\mathrm{d}t} = I_i - v_i \tag{1}$$

where v_i is the membrane potential of the ith neuron, $\tau_m = 10$ ms is the membrane time constant, and $I_i = I_{\text{Bias}} + s_i$ is the input current to neuron i. This current is composed of a constant background current $I_{\text{Bias}} = -40$ pA and synaptic currents s_i from other neurons in the network to neuron i:

$$s_i = \sum_{j=1}^{N} \omega_{ij} r_j \tag{2}$$

where N is the number of neurons. The quantity ω_{ij} is the amplitude of post-synaptic current delivered from neuron j to neuron i when neuron j spikes. The time course of the post-synaptic current is given by r_j, which evolves according to a double exponential filter:

$$\frac{\mathrm{d}r_j}{\mathrm{d}t} = -\frac{r_j}{\tau_d} + h_j$$

$$\frac{\mathrm{d}h_j}{\mathrm{d}t} = -\frac{h_j}{\tau_r} + \frac{1}{\tau_r \tau_d} \sum_{t_{jk}<t} \delta(t - t_{jk})$$

where τ_d is the decay time, τ_r is the rise time, t_{jk} is the time of the kth spike from the jth neuron, and $\delta(.)$ is the Dirac delta function. Spiking of the membrane voltage is implemented via the reset rule $v_i \rightarrow v_{\text{reset}}$ whenever $v_i > v_{\text{th}}$. A refractory period is implemented by manually holding the voltage constant at the value v_{reset} for a duration $\tau_{\text{ref}} = 2$ ms following each spike. Sample voltage activity is shown in figure 1.

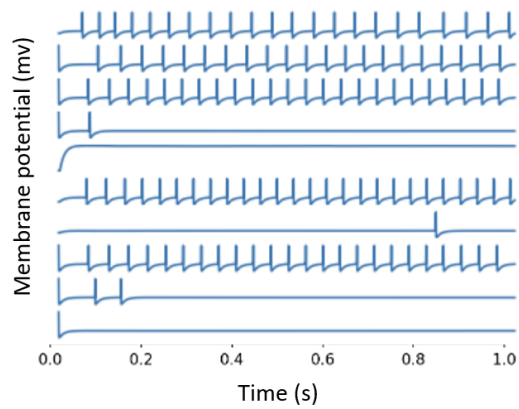

Fig. 1. Example spiking activity for a network of $N = 2048$ neurons. Only the voltage dynamics of 10 neurons are shown.

B. Locally Competitive Algorithm

The locally competitive algorithm (LCA) seeks to learn the sparse coefficients for a function library that encodes the input stimuli [4]. In our experiments the library functions are the

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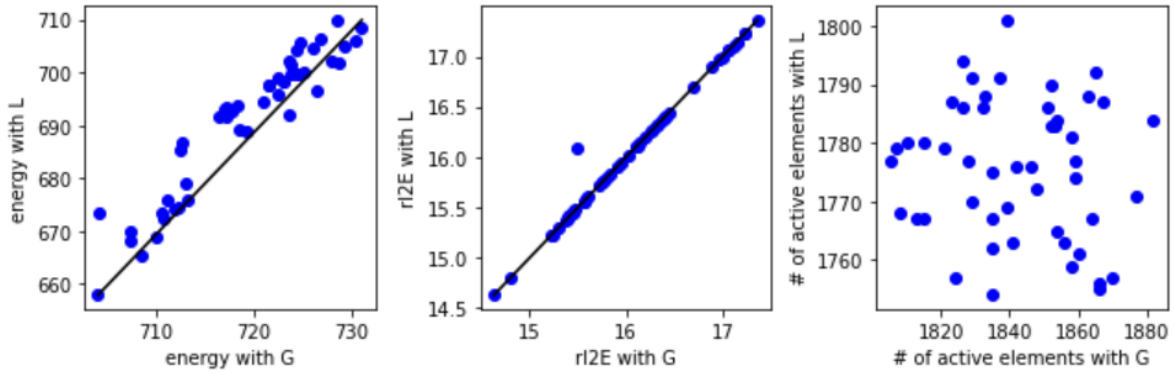

Fig. 2. Left: The energy of the original connectivity matrix G and the approximation, L. Center: The relative l^2 error of the original and approximation. Right: The number of active elements in the original versus the approximation. Neural population, N=2048

receptive fields of each of 2048 neurons. We use the same set of receptive fields used by Zhu and Rozell in [2], which were learned from the statistics of natural images. The connectivity between neurons in the network are defined based on their receptive fields based on equation 3. In this way neurons inhibit nodes with similar receptive fields.

$$\mathbf{G} = \Phi^T * \Phi \tag{3}$$

The internal states of each node is modeled as $u_m(t)$ in equation 4, which is inspired by properties observed in neural systems such as the charging of membrane potentials but in prior work [2], [4] did not incorporate spiking dynamics.

$$\dot{\mathbf{u}} = \frac{1}{\tau}(\phi^T s - \mathbf{u}(t) - (\mathbf{G} - \mathbf{I})\mathbf{a}(t)) \tag{4}$$

where a(t) is defined by a thresholding function on u.

$$\mathbf{a}(t) = \mathbf{T}_\lambda(\mathbf{u}(t)) \tag{5}$$

In order to incorporate these dynamics into a spiking network we will treat $u(t)$ as an input current for each neuron and estimate a as the firing rate of each neuron. The exact changes for each of our models is described in the following sections.

C. Spiking LCA

Our first models added in spiking dynamics with minimal changes to the models presented in [2]. In our models u was treated as an input current to each neuron and the membrane voltage dynamics were modeled with simple integrate and fire neurons. Each neurons membrane voltage was represented by v and updated with equation 6 on each time step.

$$v(t+1) = v(t) + \frac{dt}{\tau}E_L - v(t) + m * u(t)) \tag{6}$$

$$\text{if } v(t+1) > v_{threshold} \text{ then } v(t+1) = v_{reset} \tag{7}$$

E_L is the membrane's resting potential and was set to -70mV, $v_{threshold}$ is the spiking threshold for the neurons and was set to -54mV, v_{reset} was is the recovery voltage for the neurons after a spike and was set to -80mV, dt is the time step and was 1, τ is a time constant which was set to 10, and m is a scaling constant to make u a resonable curren and was set to 70. These values were taken from the original model in [2] if possible. The coefficients, a, in the spiking models were determined by estimating the firing rate of the each neuron based on the number of spikes in a small window. However, for the communication between neurons the spike train was filtered with a synaptic filter, $alpha$, defined as a decaying exponential.

$$\alpha = \frac{exp(\frac{-t}{\tau})}{\tau} \tag{8}$$

$$\tilde{\mathbf{a}} = conv(\mu, \alpha) \tag{9}$$

In equation 9 μ is the spike train over the last 32 time points and only the current time point of $\tilde{a}$ is kept to pass into the update equation for u in the LCA algorithm.

D. Neural Population Decomposition

Zhu and Rozell present several ways to decompose the connectivity matrix, G, to reflect distinct populations of excitatory and inhibitory neurons. Following their procedures we decompose the recurrent connectivity matrix $\mathbf{G}$:

$$\mathbf{G} = \mathbf{L} + \mathbf{S} \tag{10}$$

where $\mathbf{S}$ is a sparse matrix and $\mathbf{L}$ is a dense, low-rank one. They are found via the optimization:

$$\mathbf{L}, \mathbf{S} = \text{argmin}_{\mathbf{L},\mathbf{S}} ||\mathbf{L}||_* + ||\mathbf{\Lambda S}||_1 \tag{11}$$

where $\mathbf{\Lambda}$ is a diagonal matrix which promotes sparseness in $\mathbf{S}$. The operators $||.||_*$ and $||.||_1$ are the nuclear and ℓ_1-norms, respectively. The optimization 11 was solved numerically via robust principal component analysis [5].

To further, promote sparseness singular value decomposition is applied to L and only the first 110 eigenvalues are kept. To achieve this, all but the first 110 entries on the diagonal of Σ in equation 12 are set to zero.

$$\mathbf{L} = \mathbf{U}\Sigma\mathbf{V} \tag{12}$$

$$\hat{\mathbf{G}} = \hat{\mathbf{L}} + \mathbf{S} \tag{13}$$

$\hat{L}$ is the reconstructed L with only 110 eigenvalues and G is the resulting approximated G. Based on the work of Zhu and Rozell, more inhibitory dynamics can be applied to the LCA algorithm with the following equations:

$$\dot{\mathbf{u}}(\mathbf{t}) = \frac{1}{\tau}(\mathbf{\Phi}^T\mathbf{s} - (\mathbf{U}_+\mathbf{a}_{I,L1}(t) + (-\mathbf{U}_-)a_{I,Ls}(t) + \mathbf{S}_+\mathbf{a}_{I,S}(t)) + (\mathbf{I} - \mathbf{G}^{Excite})\tilde{a}(t) - \mathbf{u}(t)) \tag{14}$$

$$\dot{\mathbf{a}}_{I,L1}(t) = \frac{1}{\tau}(\Sigma\mathbf{V}_+^T\tilde{\mathbf{a}}(t) - \mathbf{a_{I,L1}}(t)) \tag{15}$$

$$\dot{\mathbf{a}}_{I,L2}(t) = \frac{1}{\tau}(\Sigma(-\mathbf{V}_-^T)\tilde{\mathbf{a}}(t) - \mathbf{a}_{I,L2}(t)) \tag{16}$$

$$\dot{\mathbf{a}}_{I,S}(t) = \frac{1}{\tau}(D\tilde{\mathbf{a}}(t) - \mathbf{a}_{I,S}(t)) \tag{17}$$

The $+$ and $-$ subscripts indicates indices containing only the positive or negative values of the original respectively. This decomposition maintain Dales Law, the assumption that each neuron is either exclusively inhibitory or excitatory.

E. More Realistic Spiking Network

After adding the very simple spiking dynamics described above, a more realistic network of Integrate and Fire neurons was implemented with methods F-I.

F. Stimuli

A visual stimulus comprised of M pixels is presented to the spiking network by appending the input current I_i in equation 1 with an additional term:

$$I_i = I_{\text{Bias}} + s_i + (\mathbf{\Phi}^\top \mathbf{s}_{\text{ext}})_i$$

where $\mathbf{s}_{\text{ext}}$ is an M-dimensional vector representing the external visual stimulus and $\mathbf{\Phi}$ is an $M \times N$ matrix representing the receptive fields of principal cells. The product $\mathbf{\Phi}^\top \mathbf{s}_{\text{ext}}$ is therefore an N-dimensional vector and the subscript i denotes its ith component. The values for the receptive fields $\mathbf{\Phi}$ that we use come from simulations of a rate-neuron model implementing sparse coding of natural images, a framework originally introduced in Ref. [6].

G. Connectivity

The matrix of synaptic connectivity is $\boldsymbol{\omega}$, whose entry in the ith row and the jth column is the term ω_{ij} first seen in equation 2. In order to integrate the spiking neuron model of section II-A with the sparse coding framework studied in Ref. [2], we let $\boldsymbol{\omega} = 0.05(\mathbf{G} - \mathbf{I}_N)/\sqrt{N}$ where $\mathbf{I}_N$ is the N-dimensional identity matrix and $\mathbf{G} = \mathbf{\Phi}^\top\mathbf{\Phi}$. The factor $0.05/\sqrt{N}$ was added to reduce the firing rates to realistic values.

H. Decomposition

For this network the decomposition in equation 11 was performed. In the results we denote as approximations, we substitute $\mathbf{L}$ for $\mathbf{G}$ in the relation $\boldsymbol{\omega} = 0.05(\mathbf{G}-\mathbf{I}_N)/\sqrt{N}$. The explicit inhibitory dynamics in equation 14 were not added.

I. Performance measures

To quantify the performance of the network in an image reconstruction task, we compute the energy function [2]:

$$E(\mathbf{a}) = \frac{1}{2}||\mathbf{s} - \mathbf{\Phi}\mathbf{a}||_2 + \lambda||\mathbf{a}||_1 \tag{18}$$

where $\mathbf{a}$ is an N-dimensional vector of neural activities. If the continuous firing rate activities $\mathbf{a}(t)$ are modeled explicitly, one may take their asymptotic values as entries to the vector in the energy calculation. Since we work with a spiking model here, we instead count spikes over a pre-specified interval of time and treat these as the entries of $\mathbf{a}$. We consider neurons that spiked at least once in this interval "active" in our results plots.

In the case that the energy above is intentionally minimized through an optimization process, the number λ is a regularization parameter controlling how much to sparsify the network activity in the solution. Since we do not directly optimize equation 18, we set $\lambda = 1$ here to compute the energy. The image reconstruction loss is equivalent to equation 18, but with the $\lambda||\mathbf{a}||_1$ term omitted.

In order to quantify the response in section III-F, we computed $||\mathbf{a}||_2$ for each orientation. The response was then normalized between 0 and 1 via the formula $(a - A_{\text{min}})/(A_{\text{max}} - A_{\text{min}})$, where a is the $||\mathbf{a}||_2$ resulting from a given orientation, A_{min} is the minimum $||\mathbf{a}||_2$ across all orientations, and A_{max} is the maximum.

III. Results and Discussion

A. Neural Spiking

Based on the model described in Methods, a spiking neural model was successfully implemented. Figure 1 shows the activity of 10 of the 2048 neurons within the spiking model. The activity is more regular than a network with randomly generated weights because it is governed by a predefined connectivity matrix (based on training done in [2]).

B. Energy Comparisons for G and L

The energy (given by Equation 18) was computed for both the original connectivity matrix, G, and the approximate low-rank matrix, L, derived from decomposition. The energy relationship between the two are approximately linear with the energy for G generally slightly more than that of L. As the energy is related to both the relative l^2 error (RL2E) and the number of active elements, the energy relationship is not as strongly linear as the RL2E relationship but not as non-linear as is the case with the number of active elements relationship. This aligns closely with the energy comparison of [2].

C. RL2E Comparison for G and L

The relative l^2 error was calculated for both the G and L matrices. The relationship between the two closely follows the identity line indicating that there was little difference in the ability to represent the stimulus in a spiking neuron network that used the original connectivity matrix or the approximate one. This aligns closely with the RL2E comparison of [2].

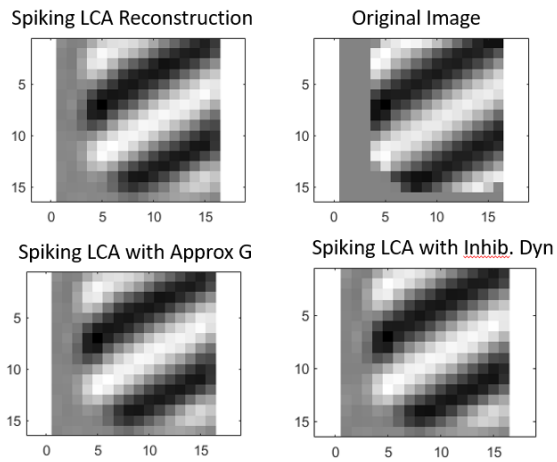

Fig. 3. Comparisons of the reconstructed images to the real image in the top right corner. All of the reconstructions look very similar, but under close inspection the reconstruction from the model with inhibitory dynamics (bottom right) is slightly worse than the other two models.

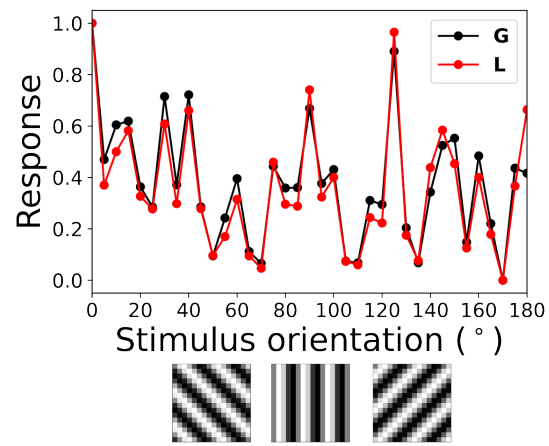

Fig. 4. Tuning curve for the spiking network (top) to static oriented grating stimuli, three examples of which are shown (bottom) for 45°, 90°, and 135°.

D. Number of Active Elements for G and L

The number of active elements were calculated for G and L to gauge the sparsity of the solution. The number of active elements in the simulations between the two were not linearly related. Additionally, the spiking model uses many more neurons than that required as compared to that of Figure 4 in [2]. This discrepancy may be due to the fact that our methodology to determine the neurons that were "active" differed from that of [2]. [2] determine that a neuron is active when the firing rate a(t) for that neuron is nonzero at the end of the simulation period. However, because we are utilizing a spiking network, we did not calculate firing rates directly. Rather, we determined that a neuron was active if it had spiked at all during the simulation. In future work, we could instead try to determine if the neuron is active by only counting spikes at the very end of simulation, thresholding the number of spikes necessary for a neuron to be considered active, or finding an accurate way to convert information about spiking patterns into the form of firing rate described in [2].

E. Image Reconstruction

In order to ensure we were successfully learning coefficients that represented the input images, a sample image was reconstructed using several of our spiking LCA models. Figure 3. The reconstructions from LCA with full connectivity matrix, G, and the approximated matrix for G from equation 13 result in identical reconstructions since the network activity does not differ significantly. However, when the inhibitory dynamics from equation 14 are added into the model the reconstruction becomes slightly worse, but still closely resembles the original.

F. Orientation Tuning

To test if our spiking network was selective for visual inputs of particular orientations, we implemented grating stimuli and computed the resulting response across a range of angles. This was done for the model with full connectivity **G** (black trace in Fig. 4), as well as the approximate sparse connectivity **L** (red trace in Fig. 4). It is evident from figure 4 that the network does not prefer one orientation over all others. Instead, it displays several local maxima and minima in its response, likely owing to the diverse orientations embedded in **G** when derived from natural stimuli. Additionally, there is little difference between the tuning curve for **G** and that for **L**. This indicates that the network connectivity responsible for selectivity to low-dimensional oriented stimuli is low-rank, even when the neurons are pulse-coupled via the spiking mechanism.

IV. Conclusion

In this work, we were able to successfully implement a spiking neural model for efficient sensory coding with a separate inhibitory interneuron population. Key results from [2] were replicated with a robust biophysical spiking model. These results include the energy, RL2E, and orientation tuning comparisions between the original connectivity matrix (G) and the low-rank approximation (L) as well as faithful image reconstruction. Future work could include developing a more robust methodology for determining the activity of a neuron in our spiking model.

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